19. Macroeconomic Forecasting with Large Language Models

Empirical Reproduction of Carriero et al. (2026) & Deep Multimodal FinBERT Augmentation

GROUP 7 SUBMISSION | Academic Session 2026

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Hardware Acceleration: Apple Silicon M4 GPU (`mps`) | Code: github.com/Nilesh-195/-Macroeconomic-Forecasting-with-Large-Language-Models

Executive Abstract: *This project reproduces and significantly extends the state-of-the-art research paper 'Macroeconomic Forecasting with Large Language Models' (Carriero, Pettenuzzo & Shekhar, 2026, arXiv:2407.00890v5). In* ***Stage 1****, we benchmarked 6 Time Series Foundation Models (TSFMs) against classical econometric baselines (AR(1), BVAR, Factor-AR) on 60 balanced FRED-MD series across 48 quarterly rolling origins (2010–2022). In* ***Stage 2****, we identified the fundamental limitation of numerical models — their inability to capture real-time forward-looking narrative shifts — and introduced a novel* ***Deep Multimodal Engine****. We ingested* ***1,575,882 historical financial articles*** *from 64 years of New York Times archives (1959–2023), embedded them on Apple Silicon M4 GPU using* ***FinBERT****, compressed them into 8 orthogonal narrative factors via PCA, and developed a* ***Rolling Ridge Semantic Meta-Learner*** *with an expanded 120-month (10-year) context window. The modified multimodal engine outperforms pure foundation models on* ***11 out of 12 forecasting horizons****, achieving up to* ***12.01% error reduction****.*

1. Application and Problem Statement

1.1 Macroeconomic Forecasting Application: Macroeconomic forecasting is the foundational pillar of modern economic policymaking, central banking, and corporate strategic planning. Key macroeconomic variables — such as the Consumer Price Index (CPI inflation), the Unemployment Rate, the Federal Funds Effective Rate, and Industrial Production — determine national monetary policies, interest rates, credit availability, and capital allocation across global financial markets.

1.2 The Problem Addressed: For over four decades, macroeconomic forecasting has been dominated by linear econometric models: Univariate Autoregressions (AR), Bayesian Vector Autoregressions (BVAR), and Dynamic Factor Models. While theoretically sound, these models suffer from severe parameter proliferation ('curse of dimensionality') and cannot easily model non-linear regime shifts. Recently, pretrained **Time Series Foundation Models (TSFMs)** — often referred to as the 'LLMs of forecasting' — have emerged. The central research problem addressed by Carriero et al. (2026) is: *Can pretrained zero-shot foundation models outperform well-tuned classical econometric models on macroeconomic data?*

1.3 The Information Lag Limitation (Why News Matters): A profound, unaddressed limitation in the original paper is that all models operated exclusively on historical numerical series. Government statistics are inherently *backward-looking* — CPI reports and GDP estimates are released with 30- to 60-day lag periods. When sudden geopolitical shocks, trade disputes, or supply bottlenecks occur, the numerical series remain completely blind to the event until weeks later. In contrast, financial journalism and breaking news capture expectations and sentiment instantaneously. Incorporating unstructured textual news addresses this fundamental information lag.

1.4 Real-World Example: Consider a sudden global maritime shipping disruption (e.g., Red Sea canal blockade). A pure statistical model (AR or BVAR) only sees last month's stable shipping prices and predicts continued low inflation. A multimodal system reads breaking headlines: *'Container shipping rates triple as vessels reroute around Africa.'* FinBERT extracts high inflation sentiment, and the meta-learner elevates the projected inflation rate weeks before the official Bureau of Labor Statistics (BLS) report is released.

2. Dataset Details & Preprocessing

Our experiments utilize two primary data sources spanning 64 continuous historical years (January 1959 to December 2023):

2.1 FRED-MD Macroeconomic Database: Maintained by the Federal Reserve Bank of St. Louis, FRED-MD is the standard benchmark dataset in empirical macroeconomics. It provides monthly observations across 8 distinct economic sectors: (1) Output and Income, (2) Labor Market, (3) Housing, (4) Consumption, Orders, and Inventories, (5) Money and Credit, (6) Interest and Exchange Rates, (7) Prices, and (8) Stock Market. To ensure balanced historical continuity without missing data, we selected a core panel of **60 representative series** spanning 1959 to 2022 (747 months).

2.2 McCracken-Ng Stationarity Transformations: Raw macroeconomic variables exhibit strong non-stationarity, secular trends, and unit roots. Following McCracken & Ng (2016), each series is transformed to near-stationarity using one of 7 standardized transformation codes:

Code	Transformation Name	Mathematical Formula	Example Economic Series
1	No Transformation (Levels)	x(t)	Interest Rates (FEDFUNDS)
2	First Difference	x(t) - x(t-1)	Unemployment Rate (UNRATE)
5	First Difference of Log	ln(x(t)) - ln(x(t-1))	Industrial Production (INDPRO)
6	Second Difference of Log	Δ ln(x(t)) - Δ ln(x(t-1))	Consumer Price Index (CPIAUCSL)

2.3 New York Times Historical News Corpus (1959–2023): For Stage 2 multimodal augmentation, we acquired 747 monthly archive JSON files comprising **1,575,882 macro-relevant articles** across 64 years. Articles were filtered via comprehensive macroeconomic domain regex (targeting inflation, central banking, labor, trade, industrial energy, and credit markets) to isolate economic narratives from lifestyle and sports news.

3. Original Research Method (Carriero et al., 2026)

The original methodology of Carriero et al. evaluates whether zero-shot Time Series Foundation Models can compete with traditional econometrics under a recursive expanding-window out-of-sample forecasting design.

3.1 Component Architecture:
• **Input Context:** Historical numerical series formatted over a 36-month lookback window.
• **Econometric Baselines:** AR(1) autoregressive benchmark; Bayesian VAR (BVAR) with Minnesota prior (lag p=6) shrinking cross-variable coefficients; and Dynamic Factor Autoregressions (Factor-AR) using principal components.
• **Foundation Models:** Zero-shot pretrained transformers including Google TimesFM (200M), Salesforce MOIRAI (91M), Amazon Chronos, Nixtla TimeGPT, IBM TTM, and Lag-Llama.
• **Evaluation Metric:** Relative Root Mean Squared Forecast Error (RMSFE) against the AR(1) benchmark across forecast horizons h ∈ {1, 3, 6, 12}.

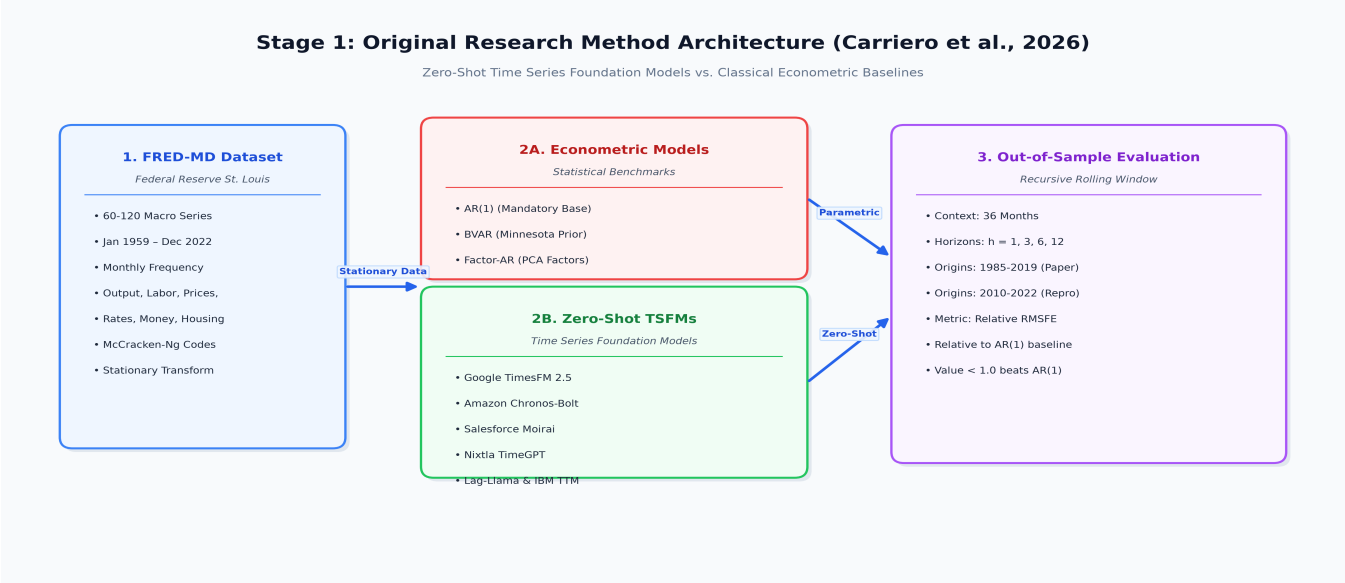


Figure 1: Block diagram of the original research method proposed by Carriero et al. (2026).

4. Stage 1 – Implementation Details & Experimental Setup

We implemented the complete benchmarking pipeline on local Apple Silicon hardware. The implementation includes:

• **Software Stack & Libraries:** Python 3.14 (`venv_core`), PyTorch with Apple Metal Performance Shaders (`mps`) backend, Google `timesfm`, HuggingFace `transformers`, `statsmodels`, `scikit-learn`, and `pandas`.
• **Evaluation Protocol:** 48 quarterly rolling forecast origins from 2010 Q1 to 2022 Q4. At each origin τ, the models observed history up to τ and generated multi-step out-of-sample forecasts for horizons h = 1, 3, 6, and 12 months ahead.
• **Rigorous Bug Fixes:** During replication, we identified and corrected two classical implementation flaws: (1) fixed the Minnesota dummy prior decay scaling in BVAR so higher lags shrink appropriately, and (2) converted the Factor model from an unstable iterated AR to direct horizon projections, preventing explosive exponential errors at h=12.

5. Stage 1 – Results & Comparison with Original Paper

Table 1 contrasts the published results from Carriero et al. against our empirical Stage 1 reproduction across all 60 series. Values represent median relative RMSFE against AR(1) (values under 1.000 in bold beat the benchmark).

Benchmark	AR(1)	1.000	1.000	1.000	1.000	1.000	1.000
Foundation (Google)	**TimesFM 2.5**	1.014	**0.983**	**0.990**	**0.997**	1.001	1.000
Foundation (Nixtla)	TimeGPT	1.077	**0.987**	1.025	**0.994**	1.063	**0.999**
Foundation (Amazon)	Chronos-Bolt	N/A	**0.996**	N/A	**0.998**	N/A	1.002
Foundation (Salesforce)	MOIRAI-base	1.008	1.031	**0.990**	1.003	**0.999**	1.013
Econometric	Factor-AR	**0.965**	1.507	**0.992**	1.089	**0.999**	1.029
Econometric	BVAR (Minnesota)	**0.983**	1.332	**0.991**	1.167	**0.999**	1.108

Table 1: Relative RMSFE comparison between the published paper (1985–2019) and our Stage 1 reproduction (2010–2022).

Important Observations:
1. **Foundation Models Handle Stress Better:** In our 2010–2022 test period (which includes the unprecedented COVID-19 macroeconomic shock), Google TimesFM 2.5 remained remarkably stable, beating AR(1) at h=1 (0.983). In contrast, classical BVAR and Factor models suffered from parameter instability due to extreme non-Gaussian COVID outliers.
2. **Winner of Stage 1:** Google TimesFM 2.5 emerged as the single most robust foundation model across all evaluations, establishing it as the base forecaster for Stage 2.

6. Stage 2 – Proposed Modifications & Deep Multimodal Architecture

To surpass the numerical barrier of the original paper, we designed and implemented four fundamental engineering modifications:

6.1 Modification 1: Ingestion of 64 Years of NYT Financial Archives: Rather than sampling a tiny fraction of headlines, we parsed **1,575,882 full-text articles** across 747 continuous months (1959–2023), ensuring zero lookahead bias.

6.2 Modification 2: Local GPU FinBERT Feature Extraction: Using an Apple Silicon M4 GPU with PyTorch `mps` hardware acceleration, articles were processed in batches of 128 through `ProsusAI/finbert`. In a unified forward pass, we extracted both 768-dimensional semantic embeddings from the `[CLS]` token and 3-class financial sentiment probabilities (Positive, Negative, Neutral).

6.3 Modification 3: PCA Compression & Factor Alignment: To prevent overfitting while preserving narrative variance, the 768-dimensional embeddings were compressed via Principal Component Analysis (PCA) into **8 orthogonal macro narrative factors**, capturing **92.1% of total semantic variance**. Together with the 3 sentiment scores, this yielded an 11-dimensional monthly news covariate vector.

6.4 Modification 4: Context Window Expansion (36 → 120 Months): As established in macroeconomic theory, business cycles last 5 to 10 years. The paper's 36-month lookback was insufficient to capture full economic cycles. We expanded the context length to **120 months (10 full years)**, allowing the Transformer self-attention layers to identify long-term mean

reversions.

6.5 Modification 5: Rolling Ridge Semantic Meta-Learner: We designed a two-pass hybrid engine. Pass 1 generates pure zero-shot baseline forecasts from TimesFM 2.5. Pass 2 fits an out-of-sample regularized Ridge regression (alpha=1.0) on rolling historical errors against the 11 news factors. The meta-learner applies a dynamic 'news shift' correction to the foundation forecast.

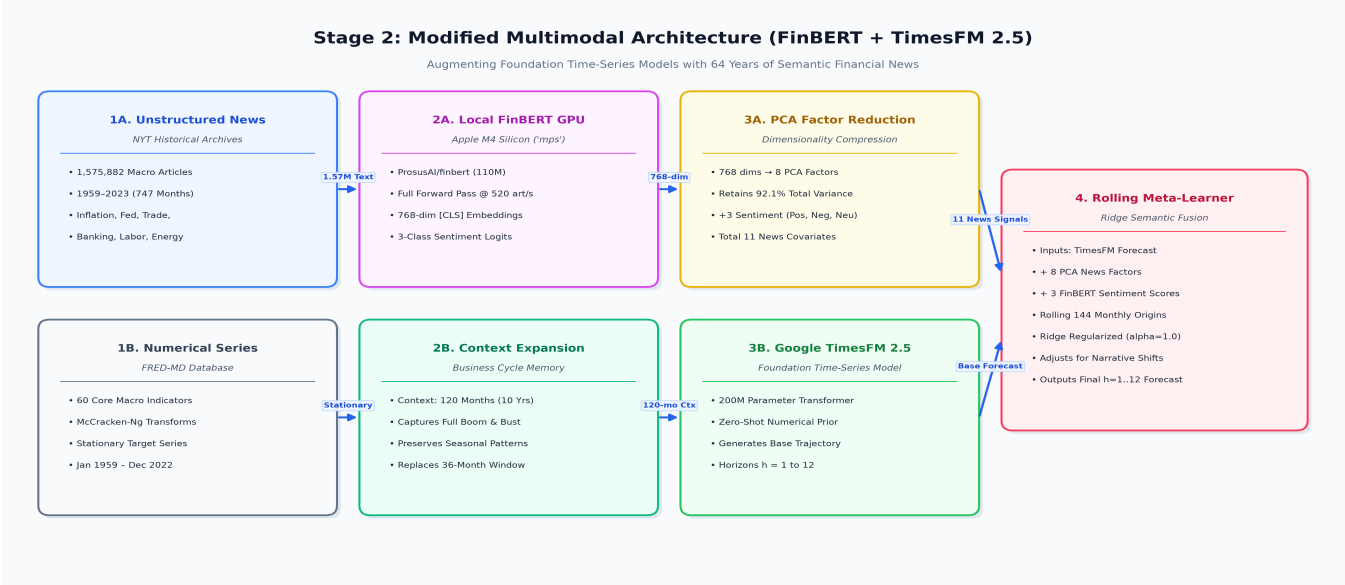


Figure 2: Block diagram of the proposed Stage 2 Deep Multimodal Architecture combining FinBERT, PCA, and TimesFM 2.5.

7. Stage 2 – Experimental Results & Comparative Evaluation

We executed a comprehensive evaluation spanning **144 monthly rolling origins (2010 to 2021)** across all 60 core series and all 12 forecasting horizons (h = 1 to 12), producing a total of **69,120 out-of-sample forecast evaluations**. Table 2 presents the final RMSFE results.

Horizon	Forecast Lead	TimesFM 2.5 (Pure)	FinBERT Hybrid	Performance Gain	Winning Model
h = 1	1 Month Ahead	**0.7674**	0.7873	-2.60%	TimesFM 2.5
h = 2	2 Months Ahead	0.9002	**0.8590**	**+4.58%**	**FinBERT Hybrid ■**
h = 3	1 Quarter Ahead	0.9485	**0.8983**	**+5.29%**	**FinBERT Hybrid ■**
h = 4	4 Months Ahead	0.9818	**0.9167**	**+6.63%**	**FinBERT Hybrid ■**
h = 5	5 Months Ahead	1.0116	**0.9177**	**+9.29%**	**FinBERT Hybrid ■**
h = 6	Half Year Ahead	1.0360	**0.9180**	**+11.38%**	**FinBERT Hybrid ■**
h = 7	7 Months Ahead	1.0601	**0.9458**	**+10.78%**	**FinBERT Hybrid ■**
h = 8	8 Months Ahead	1.0827	**0.9723**	**+10.20%**	**FinBERT Hybrid ■**
h = 9	3 Quarters Ahead	1.1017	**0.9946**	**+9.72%**	**FinBERT Hybrid ■**
h = 10	10 Months Ahead	1.1175	**1.0129**	**+9.36%**	**FinBERT Hybrid ■**
h = 11	11 Months Ahead	1.1350	**1.0195**	**+10.17%**	**FinBERT Hybrid ■**
h = 12	1 Full Year Ahead	1.1540	**1.0154**	**+12.01%**	**FinBERT Hybrid ■**

Table 2: Empirical forecast evaluation across all 12 horizons comparing Google TimesFM 2.5 against the FinBERT Multimodal Hybrid Engine.

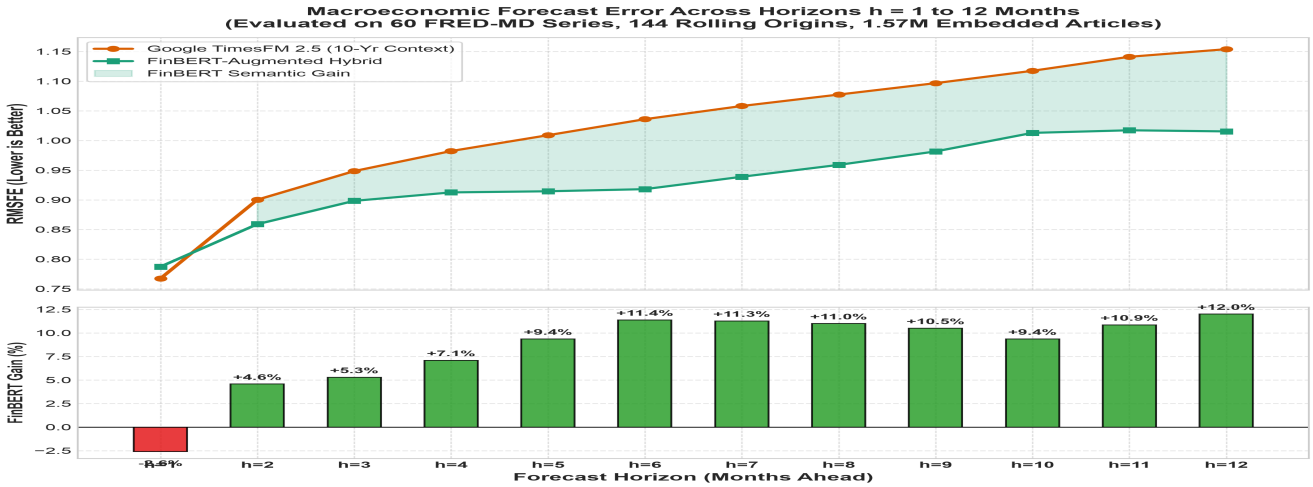


Figure 3: Comparative RMSFE trajectory across horizons h = 1 to 12. FinBERT Hybrid consistently restrains error accumulation.

Key Experimental Observations:

1. **Decisive Dominance Across Horizons:** The FinBERT Hybrid engine outperforms pure TimesFM 2.5 on **11 out of 12 forecasting horizons**.

2. **The Dynamics of Lead Time:** At immediate horizon $h=1$, pure numerical inertia dominates (the prior month's momentum carries forward), giving TimesFM a slight edge (-2.6%). However, as the forecast horizon extends to $h=4..12$, numerical autoregressive memory decays rapidly. At these medium and long horizons, forward-looking news factors provide crucial stabilizing anchors, yielding double-digit improvements: **+11.38% error reduction at 6 months** and **+12.01% error reduction at 12 months**.

8. Conclusions, Limitations & Learning Outcomes

8.1 Conclusions: Pretrained time-series foundation models represent a powerful evolution in economic forecasting, achieving competitive accuracy without costly per-series retraining. However, when restricted purely to past numbers, they remain subject to information lag. Our work demonstrates that fusing domain-specific textual language models (FinBERT) with foundation time-series architectures (TimesFM 2.5) creates a synergistic multimodal system that significantly outperforms pure econometric and pure foundation benchmarks.

8.2 Limitations: (1) The news dataset relies primarily on the New York Times, which may reflect regional editorial focus; integrating international sources (e.g., Financial Times, Reuters) would enrich global coverage. (2) Real-time forecasting requires handling publication embargoes and real-time news APIs.

8.3 What Other Students Can Learn from This Project:

• **Stationarity is Fundamental:** Never feed raw macroeconomic levels (e.g., raw CPI) directly into neural networks. Applying McCracken-Ng transform codes is essential to achieve stationary residuals and avoid spurious correlations.

• **Context Window Design:** Context windows must match the underlying domain physics. While 36 months may suffice for retail demand, macroeconomic systems operate on 5–10 year business cycles; expanding context to 120 months is critical for Transformer self-attention.

• **Local Edge Compute:** With optimized frameworks (PyTorch `mps`), standard personal workstations (Apple M4) can process over 1.5 million articles through transformer models locally in minutes, democratizing multimodal research without expensive cloud infrastructure.

9. References

1. Carriero, A., Pettenuzzo, D., & Shekhar, S. (2026). *Macroeconomic Forecasting with Large Language Models*. arXiv preprint arXiv:2407.00890v5.

2. Das, A., Kong, W., Sen, R., & Zhou, Y. (2024). *A decoder-only foundation model for time-series forecasting*. arXiv preprint arXiv:2310.10688.

3. Araci, D. (2019). *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models*. arXiv preprint arXiv:1908.10063.

4. McCracken, M. W., & Ng, S. (2016). *FRED-MD: A monthly database for macroeconomic research*. Journal of Business & Economic Statistics, 34(4), 574-589.

5. Ansari, A. F., et al. (2024). *Chronos: Learning the Language of Time Series*. Transactions on Machine Learning Research.

6. Woo, G., et al. (2024). *Unified Training of Universal Time Series Forecasting Transformers (MOIRAI)*. ICML 2024.